A Multi-Objective Framework for Prioritising Grade Separation of Railway Level Crossings

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Abstract

Railway level crossings (LCs) create a recurring conflict between road traffic and rail operations. Conventional prioritisation based primarily on Train Vehicle Units (TVU) captures exposure but does not fully measure safety risk, road-user delay, railway congestion, capacity release, reliability, accessibility, environmental effects, equity, or life-cycle cost. This paper formulates LC grade separation as a multi-objective project selection and portfolio optimisation problem. The proposed framework combines continuous road-vehicle counts, train event data, queueing simulation, safety risk modelling, railway capacity assessment, economic appraisal, and multi-objective integer programming. The methodology produces both a monetised benefit-cost appraisal and a non-monetised strategic score vector. It is designed for deployment using AI-based vehicle classification, signalling/train movement logs, GIS layers, road-network information and engineering cost estimates. The framework enables planners to identify robust priority projects and construct annual ROB/RUB/closure portfolios under capital budget constraints.

Keywords: level crossing, grade separation, ROB, RUB, TVU, multi-objective optimisation, road delay, railway congestion, safety, portfolio planning.

1 Introduction

Railway level crossings are among the most visible points of interaction between the railway system and the road network. Their elimination through Road Over Bridges (ROBs), Road Under Bridges (RUBs), diversion roads, gate closure, or other grade-separation measures can yield safety, road-mobility and railway-capacity benefits. However, the prioritisation problem is not trivial. A gate with high road traffic may impose large road-user delay but modest railway consequences. Another gate may have lower road traffic but lie on a saturated railway corridor where elimination improves pathing flexibility. A third gate may have modest traffic but poor sighting, unsafe approach geometry, or high community severance.

Traditional prioritisation often begins with traffic exposure indicators such as Train Vehicle Units (TVU), typically defined as the product of daily train units and daily road vehicle units:

$$TVU_i = T_i V_i, \tag{1}$$

where T_i is the number of train movements and V_i is the number of road vehicles at level crossing i over a representative day. TVU is an important exposure measure, but it is not itself a welfare measure. Two gates may have the same TVU but impose very different social costs. For example, 40 trains and 3000 vehicles per day produce the same TVU as 120 trains and 1000 vehicles per day, but the first may mainly create road queues while the second may create railway operational constraints.

This paper proposes that LC elimination should be formulated as a multi-objective portfolio problem. The central question is not simply:

> Which LC has the highest TVU?

but rather:

> Under a constrained annual capital budget, which portfolio of LC interventions provides the best combination of safety, road mobility, railway capacity, reliability, accessibility, equity, environmental benefit and economic value?

2 Problem Definition

Let $\mathcal{I}$ denote the set of candidate LCs and $\mathcal{K}_i$ denote the feasible treatment alternatives for LC i, such as retain, ROB, RUB, closure, diversion or limited improvement. Define the binary decision variable

$$x_{ik} = \begin{cases} 1, & \text{if treatment } k \in \mathcal{K}_i \text{ is selected for LC } i, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

Each LC receives exactly one treatment:

$$\sum_{k \in \mathcal{K}_i} x_{ik} = 1 \qquad \forall i \in \mathcal{I}. \tag{3}$$

Let C_{ik} be the present value of life-cycle cost for treatment k at LC i, and let B be the available capital budget. The budget constraint is

$$\sum_{i \in \mathcal{I}} \sum_{k \in \mathcal{K}_i} C_{ik} x_{ik} \leq B. \tag{4}$$

The benefits of treatment k at LC i are represented by a vector

$$\mathbf{b}_{ik} = [S_{ik}, R_{ik}, G_{ik}, K_{ik}, Q_{ik}, A_{ik}, E_{ik}, H_{ik}, -C_{ik}]^\top, \tag{5}$$

where S is safety benefit, R road-user delay saving, G railway delay saving, K railway capacity benefit, Q reliability improvement, A accessibility gain, E environmental benefit, H equity or distributional benefit, and C cost. The objective is not one scalar function but a vector optimisation problem:

$$\max_{x} \sum_{i \in \mathcal{I}} \sum_{k \in \mathcal{K}_i} \mathbf{b}_{ik} x_{ik} \tag{6}$$

subject to budget, engineering, geographic and policy constraints.

3 Data Architecture

The proposed framework requires a common LC master table. Table 1 summarises the principal data sources and derived variables.

Table 1: Data architecture for LC grade-separation prioritisation.

Data block	Example inputs	Derived measures
LC inventory	Gate number, location, section, type, manning, interlocking, road class	Base exposure, operational classification, treatment feasibility
Train data	Train movements, block occupation, signalling logs, timetable, data logger events	Gate closure events, train interaction, capacity constraints
Road traffic	AI vehicle counts at 10-second intervals, vehicle class, turning movements	Arrival profiles, queue growth, person delay, truck delay
Safety	Accident history, near-miss logs, visibility, approach geometry, road speed, train speed	Expected accident risk, severity-weighted risk reduction
Engineering	Alignment, land, utilities, drainage, vertical clearance, geotechnical conditions	ROB/RUB/closure feasibility and cost
Network GIS	Roads, settlements, schools, hospitals, markets, stations, detour routes	Accessibility, severance, diversion, network impact
Socioeconomic	Population, vulnerable groups, income proxies, settlement type	Equity-weighted benefits and distributional reporting
Environment	Emission factors, vehicle classes, queue/idling profiles	Fuel, CO2, PM, NOx changes

The most important methodological shift is the replacement of short-period manual traffic counts with continuous AI-based observations. If vehicle counts are available at interval $\Delta t = 10$ seconds, let $q_{ic}(t)$ be the arrival rate of class c at LC i. Combined with train passage and gate-closure times, this allows reconstruction of the road queue process over the day.

4 Road Delay Model

For each closure event g at LC i, let t_g^{close} and t_g^{open} denote closure and reopening times. Let $A_{ic}(t)$ be arrivals of vehicle class c during interval t, and $D_{ic}(t)$ be departures when the gate is open. Queue evolution may be written as

$$Q_{ic}(t + \Delta t) = \max\{0, Q_{ic}(t) + A_{ic}(t) - D_{ic}(t)\}. \tag{7}$$

The total vehicle delay is

$$VD_i = \sum_c \int_0^{24h} Q_{ic}(t)\, dt. \tag{8}$$

The total person delay is

$$PD_i = \sum_c \int_0^{24h} Q_{ic}(t)\, Occ_c\, dt, \tag{9}$$

where Occ_c is vehicle occupancy for class c. The monetised annual road benefit from full grade separation is

$$B_i^{road} = 365 \left[\sum_c PD_{ic} VOT_c + \sum_{c \in \mathcal{F}} VD_{ic} VOC_c^{time} \right], \tag{10}$$

where VOT_c is the value of time and VOC_c^{time} is time-related vehicle operating cost for freight vehicle classes $\mathcal{F}$.

5 Railway Delay and Capacity Model

LCs may affect rail operations through gate working, signal clearance, approach control, speed restrictions, timetable padding, train regulation and section capacity. Let $Delay_{ir}$ be the delay imposed on train r due to LC interaction. Direct train delay cost is

$$B_i^{rail-delay} = \sum_r Delay_{ir} \cdot VOT_r^{train}. \tag{11}$$

However, rail capacity benefits can be more important than observed delay. Let K_s^0 and K_s^1 denote sectional capacity before and after grade separation on section s containing LC i. Then

$$\Delta K_{is} = K_s^1 - K_s^0. \tag{12}$$

If the section is congested, the economic value of additional paths may be approximated by

$$B_i^{capacity} = \lambda_s \Delta K_{is}, \tag{13}$$

where λ_s is the shadow price of an additional train path on section s. This shadow price may be estimated from timetable conflicts, denied demand, freight waiting time, or network simulation.

6 Safety Risk Model

Expected annual accident cost at LC i can be represented as

$$SC_i = \sum_a P_{ia} L_{ia}, \tag{14}$$

where a indexes accident types, P_{ia} is the probability of accident type a, and L_{ia} is the expected consequence. A statistical risk model can be estimated as

$$\begin{aligned} P(Accident_i = 1) = F(&\alpha_0 + \alpha_1 \log(1 + TVU_i) + \alpha_2 Geom_i \\ &n + \alpha_3 Vis_i + \alpha_4 Speed_i + \alpha_5 GateType_i + \cdots). \end{aligned} \tag{15}$$

The safety benefit of a treatment is the expected reduction in risk:

$$S_{ik} = SC_i^0 - SC_{ik}^1. \tag{16}$$

In decision support, safety should be reported both in monetised form and as physical risk reduction, such as expected fatalities, serious injuries and incidents avoided.

7 Reliability, Accessibility, Environment and Equity

Mean delay is an incomplete measure. Reliability can be quantified by high-percentile delay or exceedance probabilities:

$$Q_i = P(D_i^{road} > \tau_R) + P(D_i^{rail} > \tau_G), \tag{17}$$

where τ_R and τ_G are road and rail delay thresholds.

Accessibility benefits can be measured for settlements near the LC. For zone z, define Hansen-type accessibility to opportunities O_j:

$$A_z = \sum_j O_j \exp(-\lambda t_{zj}). \tag{18}$$

The intervention benefit is

$$\Delta A_z = A_z^1 - A_z^0. \tag{19}$$

This captures reduced severance and improved access to schools, hospitals, markets, stations and employment.

Environmental benefits arise from reduced idling and stop-start movement:

$$E_i = \sum_c \left(Fuel_{ic}^0 - Fuel_{ic}^1\right) EF_c, \tag{20}$$

where EF_c is an emission factor. Care is required where improved roads induce additional traffic.

Equity is handled by reporting benefit distribution by population group q:

$$H_i = \left[Benefit_{i1}, Benefit_{i2}, \ldots, Benefit_{iQ}\right]. \tag{21}$$

If desired, explicit policy weights ω_q can be applied:

$$H_i^{weighted} = \sum_q \omega_q Benefit_{iq}. \tag{22}$$

The weights should be transparent because they are social choices, not engineering constants.

8 Economic Appraisal

The present value of benefits for treatment k at LC i is

$$PV(B_{ik}) = \sum_{t=1}^{T} \frac{B_{ikt}}{(1+r)^t}, \tag{23}$$

where r is the social discount rate. The life-cycle cost is

$$PV(C_{ik}) = C_{ik}^{capital} + \sum_{t=1}^{T} \frac{M_{ikt}}{(1+r)^t}. \tag{24}$$

Define the Value of Grade Separation as

$$VGS_{ik} = PV(B_{ik}^{safety} + B_{ik}^{road} + B_{ik}^{rail-delay} + B_{ik}^{capacity} + B_{ik}^{reliability} + B_{ik}^{environment} + B_{ik}^{accessibility}). \tag{25}$$

Then

$$NPV_{ik} = VGS_{ik} - PV(C_{ik}), \tag{26}$$

$$BCR_{ik} = \frac{VGS_{ik}}{PV(C_{ik})}. \tag{27}$$

These economic indicators are useful, but they should be reported with the non-monetised score vector so that safety, equity, rail capacity and strategic objectives are not hidden inside a single number.

9 Multi-Objective Optimisation

A pure weighted-sum formulation is

$$\max_{x} \sum_{i,k} \left(\mathbf{w}^{\top} \mathbf{z}_{ik} \right) x_{ik}, \tag{28}$$

where $\mathbf{z}_{ik}$ is the normalised benefit vector and $\mathbf{w}$ is a vector of policy weights. However, choosing $\mathbf{w}$ is a policy decision. Therefore, the recommended workflow is:

1. Estimate the full benefit vector for every feasible LC treatment.
2. Identify Pareto-efficient projects and project portfolios.
3. Generate alternative portfolios under different transparent weighting schemes.
4. Identify robust projects that appear across many plausible portfolios.
5. Present trade-offs explicitly to the decision-maker.

A constrained optimisation formulation can be written as

$$\max_{x} \quad \sum_{i,k} U_{ik} x_{ik} \tag{29}$$

$$\text{s.t.} \quad \sum_{i,k} C_{ik} x_{ik} \leq B, \tag{30}$$

$$\sum_{i,k} S_{ik} x_{ik} \geq S_{min}, \tag{31}$$

$$\sum_{i \in \mathcal{R}, k} x_{ik} \geq N_{rural}, \tag{32}$$

$$\sum_{i \in \mathcal{C}, k} x_{ik} \geq N_{congested}, \tag{33}$$

$$\sum_{k \in \mathcal{K}_i} x_{ik} = 1 \quad \forall i, \tag{34}$$

$$x_{ik} \in \{0, 1\}. \tag{35}$$

Here U_{ik} may be a weighted utility score, $\mathcal{R}$ is a set of rural or socially important LCs, and $\mathcal{C}$ is a set of LCs on congested rail corridors.

10 Implementation Workflow

The framework can be implemented in seven stages.

Stage 1: Master database creation. Build a geocoded LC inventory with engineering attributes, road class, railway section, train counts and treatment feasibility.

Stage 2: Continuous traffic measurement. Deploy AI-based vehicle counting and classification, producing 10-second arrival profiles by vehicle class.

Stage 3: Train-event integration. Link train movements, gate-closure events, signalling logs and data logger records.

Stage 4: Queue and delay simulation. Estimate road queues, vehicle delay, person delay and reliability indicators.

Stage 5: Rail-capacity assessment. Estimate train delay, section-capacity release and shadow value of train paths.

Stage 6: Economic and strategic appraisal. Calculate safety, road, rail, accessibility, environmental, equity and cost indicators.

Stage 7: Portfolio optimisation. Generate Pareto-efficient and budget-constrained ROB/RUB/closure portfolios.

11 Illustrative Formula-Based Example

Consider an LC with 40 trains per day and 3000 road vehicles per day. Suppose the average closure duration is 6 minutes, the average arrival rate during closures is 2.1 vehicles per minute, and average queue dissipation takes 4 minutes after opening. A simple triangular approximation of vehicle delay per closure is

$$VD_g \approx \frac{1}{2} Q_g (T_g + D_g), \tag{36}$$

where Q_g is maximum queue, T_g is closure time and D_g is dissipation time. If

$$Q_g = 2.1 \times 6 = 12.6 \text{ vehicles}, \tag{37}$$

then

$$VD_g \approx \frac{1}{2} \times 12.6 \times (6 + 4) = 63 \text{ vehicle-minutes}. \tag{38}$$

For 40 closures per day,

$$VD_{daily} = 2520 \text{ vehicle-minutes} = 42 \text{ vehicle-hours}. \tag{39}$$

This simple example ignores peak bunching and class-specific queues. With 10-second AI counts, the actual queue profile can be reconstructed, allowing much more accurate estimation of benefits.

12 Discussion

The proposed approach changes the role of TVU. TVU remains a useful exposure measure, but it should not be the dominant objective where richer data exist. Continuous road-traffic

counts and railway event data allow direct estimation of queueing, delay, reliability and rail operational effects. GIS and socioeconomic data allow assessment of accessibility, severance and equity. Engineering databases allow more realistic treatment-specific cost estimation.

The framework also clarifies the distinction between technical analysis and policy choice. The technical system estimates consequences. The policymaker decides trade-offs between safety, congestion, capacity, equity, accessibility and cost. This is more transparent than embedding unobserved weights in a single ranking formula.

13 Conclusion

LC grade separation is best treated as a multi-objective infrastructure portfolio problem. A robust prioritisation system should combine safety risk, road-user delay, railway delay, rail-capacity release, reliability, accessibility, environment, equity and life-cycle cost. The proposed framework provides equations, data architecture and optimisation logic for such a system. It is immediately compatible with AI-based vehicle-counting systems, railway signalling/train logs, GIS databases and standard economic appraisal. The result is a decision-support system that can rank individual LCs, select annual project portfolios, and transparently show how priorities change when policy objectives change.

References

[1] Ben-Akiva, M. and Lerman, S. R. (1985). *Discrete Choice Analysis: Theory and Application to Travel Demand*. MIT Press.

[2] Ortuzar, J. de D. and Willumsen, L. G. (2011). *Modelling Transport*. Wiley.

[3] Litman, T. (2024). *Transportation Cost and Benefit Analysis*. Victoria Transport Policy Institute.

[4] Miettinen, K. (1999). *Nonlinear Multiobjective Optimization*. Springer.

[5] Nemhauser, G. L. and Wolsey, L. A. (1988). *Integer and Combinatorial Optimization*. Wiley.